

Analyzing attacks using the Exchange vulnerabilities CVE-2022-41040 and CVE-2022-41082

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Content

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Threats intelligence

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November 8, 2022 update – Microsoft has [released patches](#) for these issues. While Microsoft has not seen any further exploitation of these vulnerabilities in the wild since the targeted use in August, it is highly recommended that organizations patch their systems as attackers often reverse engineer patches to develop exploits.

October 1, 2022 update – Added information about *Exploit:Script/ExchgProxyRequest.A*, Microsoft Defender AV’s robust detection for exploit behavior related to this threat. We also removed a section on MFA as a mitigation, which was included in a prior version of this blog as standard guidance.

Microsoft is aware of [limited targeted attacks](#) using two reported zero-day vulnerabilities affecting Microsoft Exchange Server 2013, Exchange Server 2016, and Exchange Server 2019. The first one, identified as CVE-2022-41040, is a server-side request forgery (SSRF) vulnerability, while the second one, identified as CVE-2022-41082, allows remote code execution (RCE) when Exchange PowerShell is accessible to the attacker. Refer to the [Microsoft Security Response Center blog](#) for mitigation guidance regarding these vulnerabilities.

CVE-2022-41040 can enable an authenticated attacker to remotely trigger CVE-2022-41082. However, authenticated access to the vulnerable Exchange Server is necessary to successfully exploit either vulnerability, and they can be used separately.

Microsoft released patches for these issues on November 8, 2022. Customers who haven’t patched yet are urged to do so as soon as possible. Mitigation guidance is still provided here for organizations that have not yet deployed a mitigation, and can be used while deploying patches. Customers are encouraged to enable the [Exchange Emergency Mitigation on Service](#), which allows mitigations to be deployed automatically for future incidents.

Microsoft Defender Antivirus and Microsoft Defender for Endpoint detect malware and activity associated with these attacks. Microsoft will continue to monitor threats that take advantage of these vulnerabilities and take necessary response actions to protect customers.

Analysis of observed activity

Attacks using Exchange vulnerabilities prior to public disclosure

MSTIC observed activity related to a single activity group in August 2022 that achieved initial access and compromised Exchange servers by chaining CVE-2022-41040 and CVE-2022-41082 in a small number of targeted attacks. These attacks installed the Chopper web shell to facilitate hands-on-keyboard access, which the attackers used to perform Active Directory reconnaissance and data exfiltration. Microsoft observed these attacks in fewer than 10 organizations globally. MSTIC assesses with medium confidence that the single activity group is likely to be a state-sponsored organization.

Microsoft researchers were investigating these attacks to determine if there was a new exploitation vector in Exchange involved when the Zero Day Initiative (ZDI) disclosed CVE-2022-41040 and CVE-2022-41082 to Microsoft Security Response Center (MSRC) in September 2022.

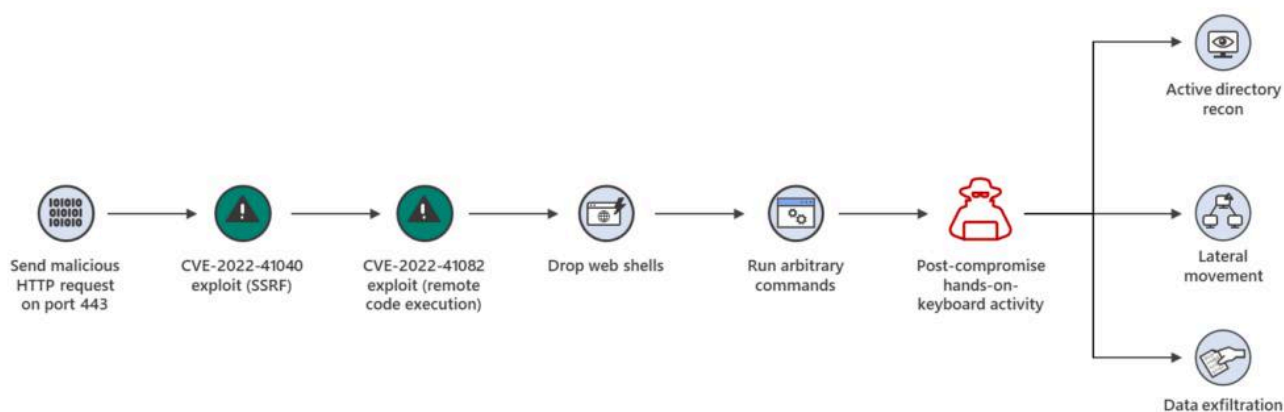


Figure 1: Diagram of attacks using Exchange vulnerabilities CVE-2022-41040 and CVE-2022-41082

Observed activity after public disclosure

On September 28, 2022, GTSC released a [blog](#) disclosing an exploit previously reported to Microsoft via the Zero Day Initiative and detailing its use in an attack in the wild. Their blog details one example of chained exploitation of CVE-2022-41040 and CVE-2022-41082 and discusses the exploitation details of CVE-2022-41040. It is expected that similar threats and overall exploitation of these vulnerabilities will increase, as security researchers and cybercriminals adopt the published research into their toolkits and proof of concept code becomes available.

While these vulnerabilities require authentication, the authentication needed for exploitation can be that of a standard user. Standard user credentials can be acquired via many different attacks, such as password spray or purchase via the cybercriminal economy. Prior Exchange vulnerabilities that require authentication have been adopted into the toolkits of attackers who deploy ransomware, and these vulnerabilities are likely to be included in similar attacks due to the highly privileged access Exchange systems confer onto an attacker.

Mitigation

Customers should refer to [Microsoft Security Response Center's post](#) for the latest on mitigations for the Exchange product.

Microsoft Exchange Server customers using [Microsoft 365 Defender](#) are advised to follow this checklist:

- Turn on [cloud-delivered protection](#) in Microsoft Defender Antivirus or the equivalent for your antivirus product to cover rapidly evolving attacker tools and techniques. Cloud-

based machine learning protections block a huge majority of new and unknown variants.

- Turn on [tamper protection](#) features to prevent attackers from stopping security services.
- Run [EDR in block mode](#) so that Microsoft Defender for Endpoint can block malicious artifacts, even when your non-Microsoft antivirus doesn't detect the threat or when Microsoft Defender Antivirus is running in passive mode. EDR in block mode works behind the scenes to remediate malicious artifacts that are detected post-breach.
- Enable [network protection](#) to prevent applications or users from accessing malicious domains and other malicious content on the internet.
- Enable [investigation and remediation](#) in full automated mode to allow Microsoft Defender for Endpoint to take immediate action on alerts to resolve breaches, significantly reducing alert volume.
- Use [device discovery](#) to increase your visibility into your network by finding unmanaged devices on your network and onboarding them to Microsoft Defender for Endpoint.

Detection

Microsoft Defender Antivirus

Microsoft Exchange AMSI integration and Antivirus Exclusions

Exchange supports the integration with the Antimalware Scan Interface (AMSI) since the June 2021 Quarterly Updates for Exchange. It is highly recommended to ensure these updates are installed and AMSI is working using the [guidance provided by the Exchange Team](#), as this integration provides the best ability for Defender Antivirus to detect and block exploitation of vulnerabilities on Exchange.

Many organizations exclude Exchange directories from antivirus scans for performance reasons. It's highly recommended to audit AV exclusions on Exchange systems and assess if they can be removed without impacting performance and still ensure the highest level of protection. Exclusions can be managed via Group Policy, PowerShell, or systems management tools like System Center Configuration Manager.

To audit AV exclusions on an Exchange Server running Defender Antivirus, launch the `Get-MpPreference` command from an elevated PowerShell prompt.

If exclusions cannot be removed for Exchange processes and folders, running Quick Scan in Defender Antivirus scans Exchange directories and files regardless of exclusions.

Microsoft Defender Antivirus detects the post-exploitation malware currently used in-the-wild exploitation of this vulnerability as the following:

| **Microsoft Defender Antivirus detections** | **MITRE **ATT&CK Tactics observed ** | | |
[Exploit:Script/ExchgProxyRequest.A](#) [Exploit:Script/ExchgProxyRequest.B](#) [Exploit:Script/ExchgProxyRequest.C](#) (the most robust defense from Microsoft Defender AV against this threat; requires Exchange AMSI to be enabled) | Initial Access | | | [Backdoor:ASP/Webshell.Y](#) |
Persistence | | | [Backdoor:Win32/RewriteHttp.A](#) | Persistence | | | [Backdoor:JS/SimChocexSh](#)

[ell.A!dha](#) | Persistence | | | [Behavior:Win32/IISExchgDropWebshell.A!dha](#) | Persistence | | | [Behavior:Win32/IISExchgDropWebshell.A](#) | Persistence | | | [Trojan:Win32/IISExchgSpawnCMD.A](#) | Execution | | | [Trojan:Win32/WebShellTerminal.A](#) | Execution | | | [Trojan:Win32/WebShellTerminal.B](#) | Execution | |

Microsoft Defender for Endpoint

[Microsoft Defender for Endpoint](#) detects post-exploitation activity. The following alerts could be related to this threat:

| **Indicators of attack*** | ****MITRE** ****ATT&CK Tactics observed** * | | | Possible web shell installation | Persistence | | | Possible IIS web shell | Persistence | | | Suspicious Exchange Process Execution | Execution | | | Possible exploitation of Exchange Server vulnerabilities (Requires Exchange AMSI to be enabled) | Initial Access | | | Suspicious processes indicative of a web shell | Persistence | | | Possible IIS compromise | Initial Access | |

As of this writing, Defender for Endpoint customers with Microsoft Defender Antivirus enabled can also detect the web shell malware used in in-the-wild exploitation of this vulnerability with the following alerts:

| **Indicators of attack**** | ***MITRE ATT&CK Tactics observed** * | | | 'Chopper' malware was detected on an IIS Web server | Persistence | | | 'Chopper' high-severity malware was detected | Persistence | |

Microsoft Defender Threat Intelligence

[Microsoft Defender Threat Intelligence](#) (MDTI) maps the internet to expose threat actors and their infrastructure. As indicators of compromise (IOCs) associated with threat actors targeting the vulnerabilities described in this writeup are surfaced, Microsoft Defender Threat Intelligence Community members and customers can find summary and enrichment information for all IOCs within the Microsoft Defender Threat Intelligence portal.

Microsoft Defender Vulnerability Management

Microsoft Defender Vulnerability Management identifies devices in an associated tenant environment that might be affected by CVE-2022-41040 and CVE-2022-41082. These vulnerabilities have been added to the CISA known exploited vulnerabilities list and are considered in the overall organizational [exposure score](#). Customers can use the following capabilities to identify vulnerable devices and assess exposure:

- Use the dedicated dashboard for each of CVE-2022-41040 and CVE-2022-41082 to get a consolidated view of various findings across vulnerable devices and software.
- Use the *DeviceTvmSoftwareVulnerabilities* table in advanced hunting to identify vulnerabilities in installed software on devices. Refer to the following query to run:

```
DeviceTvmSoftwareVulnerabilities
| where CveId in ("CVE-2022-41040", "CVE-2022-41082")
```

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CVE-2022-41040

Zero-day

[Open vulnerability page](#)
[Report inaccuracy](#)

[Vulnerability details](#)
[Exposed devices](#)
[Related software](#)

① Includes a zero-day vulnerability, which is a publicly disclosed vulnerability for which no official patches or security updates have been released. Zero-day vulnerabilities often have high severity levels and are actively exploited.

① Legal Notice The vulnerability data provided and shown as part of your Microsoft Defender for Endpoint (MDE) services is made available to you ...

Vulnerability description

Microsoft Exchange Server Elevation of Privilege Vulnerability

Vulnerability details

Vulnerability name CVE-2022-41040	Severity High
CVSS 8.8	CVSS Version 3
Published on 9/29/2022	Updated on 9/30/2022
Age 3 days	Related software Microsoft Exchange Server 2013 (+ 2 more)

Threat insights

Public Yes	Verified Yes
Exploit kits No	Type Remote

Reference

- <https://msrc-blog.microsoft.com/2022/09/29/customer-guidance-for-reported-zero-day-vulnerabilities-in-microsoft-exchange-server>
- <https://www.cisa.gov/known-exploited-vulnerabilities-catalog>

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Figure 2: Screenshot of the CVE information page where users can also take a look at related exposed device, software information, open vulnerability page, report inaccuracy, or read other useful references.

NOTE: The assessments above do not currently account for the existence of a workaround mitigation on the device. Microsoft will continue to improve these capabilities based on the latest information from the threat landscape.

Advanced hunting

Microsoft Sentinel

Based on what we're seeing in the wild, Microsoft Sentinel customers can use the following techniques for web shell-related attacks connected to these vulnerabilities. Our post on [web shell threat hunting with Microsoft Sentinel](#) also provides guidance on looking for web shells in general.

The [Exchange SSRF Autodiscover ProxyShell](#) detection, which was created in response to ProxyShell, can be used for queries due to functional similarities with this threat. Also, the new [Exchange Server Suspicious File Downloads](#) and [Exchange Worker Process Making Remote Call](#) queries specifically look for suspicious downloads or activity in IIS logs. In addition to these, we have a few more that could be helpful in looking for post-exploitation activity:

- [Exchange OAB virtual directory attribute containing potential web shell](#)
- [Web shell activity](#)
- [Malicious web application requests linked with Microsoft Defender for Endpoint alerts](#)
- [Exchange IIS worker dropping web shell](#)
- [Web shell detection](#)

Microsoft 365 Defender

To locate related activity, Microsoft 365 Defender customers can run the following advanced hunting queries:

Chopper web shell

Use this query to hunt for Chopper web shell activity:

```
DeviceProcessEvents
| where InitiatingProcessFileName =~ "w3wp.exe"
| where ProcessCommandLine has_any ("&ipconfig&echo", "&quser&echo", "&whoami&echo",
"&c:&echo", "&cd&echo", "&dir&echo", "&echo [E]", "&echo [S]")
```

Suspicious files in Exchange directories

Use this query to hunt for suspicious files in Exchange directories:

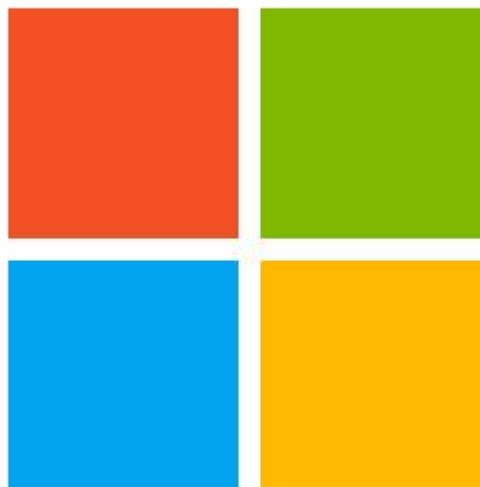
```
DeviceFileEvents
| where Timestamp >= ago(7d)
| where InitiatingProcessFileName == "w3wp.exe"
| where FolderPath has "FrontEnd\\HttpProxy\\"
| where InitiatingProcessCommandLine contains "MSExchange"
| project FileName, FolderPath, SHA256, InitiatingProcessCommandLine, DeviceId, Timestamp
```

External attack surface management

Microsoft Defender External Attack Surface Management

[Microsoft Defender External Attack Surface Management](#) continuously discovers and maps your digital attack surface to provide an external view of your online infrastructure. Attack Surface Insights are generated by leveraging vulnerability and infrastructure data to showcase the key areas of concern for your organization.

A High Severity Observation has been published to surface assets within an attack surface which should be examined for application of the mitigation steps described above. This insight, titled *CVE-2022-41082 & CVE-2022-41040 - Microsoft Exchange Server Authenticated SSRF and PowerShell RCE*, can be found under the high severity observations section of the Attack Surface Summary dashboard.



Microsoft Threat Intelligence

[See Microsoft Threat Intelligence posts](#)

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July 27 5 min read

Enhancing AI security through global AI red teaming

Microsoft's External Red Team Alliance (EXTRA) is a global AI security initiative designed to advance AI safety research and red teaming.

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